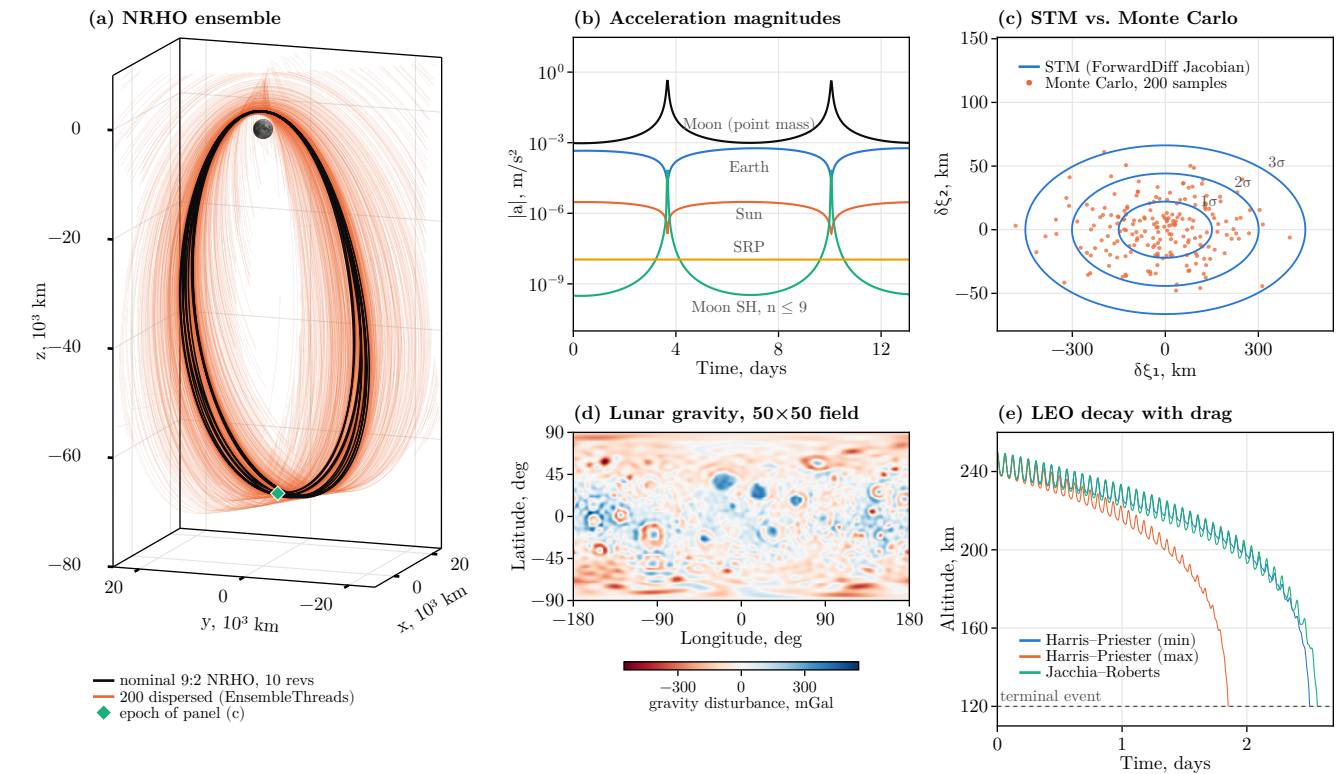

(a) NRHO ensemble
z, 10³ km
y, 10³ km
x, 10³ km
nominal 9:2 NRHO, 10 revs
200 dispersed (EnsembleThreads)
epoch of panel (c)
(b) Acceleration magnitudes
|a|, m/s²
Moon (point mass)
Earth
Sun
SRP
Moon SH, n ≤ 9
Time, days
(c) STM vs. Monte Carlo
STM (ForwardDiff Jacobian)
Monte Carlo, 200 samples
3σ
2σ
1σ
δξ2, km
δξ1, km
(d) Lunar gravity, 50×50 field
Latitude, deg
Longitude, deg
gravity disturbance, mGal
(e) LEO decay with drag
Altitude, km
Harris–Priester (min)
Harris–Priester (max)
Jacchia–Roberts
terminal event
Time, days